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IMPLEMENT YOUR OWN PING PROGRAM

AIM:

To implement a Ping program using Python to check host connectivity and calculate round-trip time (RTT).

ALGORITHM:

1. Read the destination hostname or IP address from the user.
2. Create a raw ICMP socket.
3. Calculate the checksum for the ICMP packet.
4. Construct an ICMP Echo Request packet.
5. Send the packet to the destination host.
6. Receive the Echo Reply from the destination.
7. Calculate the round-trip time (RTT) in milliseconds.
8. Display the response or a timeout message.
9. Repeat the process four times.
10. Close the socket after each request.

PROGRAM:

```
import socket
import struct
import time
import os

def checksum(data):
 total = 0

for i in range(0, len(data) - 1, 2):
 total += (data[i]> 16) + (total & 0xffff)
 total += total >> 16

return (~total) & 0xffff

def ping(host):
 sock = None
```

```
try:
 addr = socket.gethostbyname(host)

sock = socket.socket(
 socket.AF_INET,
 socket.SOCK_RAW,
 socket.getprotobyname("icmp")
 )

sock.settimeout(2)

pid = os.getpid() & 0xffff
 data = b"Computer Networks Ping"

header = struct.pack("!BBHHH", 8, 0, 0, pid, 1)
 packet = header + data

check = checksum(packet)

packet = struct.pack(
 "!BBHHH", 8, 0, check, pid, 1
 ) + data

start = time.time()
 sock.sendto(packet, (addr, 1))

reply, address = sock.recvfrom(1024)

rtt = (time.time() - start) * 1000

print("Reply from", address[0],
 ": time =", round(rtt, 2), "ms")

except socket.timeout:
 print("Request timed out")

except PermissionError:
 print("Run with administrator/root privileges")

except OSError as e:
 print("Error:", e)
```

```
finally:
 if sock is not None:
 sock.close()

host = input("Enter hostname or IP address: ")

for i in range(4):
 print("Ping attempt", i + 1)
 ping(host)
 time.sleep(1)
```

OUTPUT:

```
Enter hostname or IP address: 8.8.8.8

Ping attempt 1

Reply from 8.8.8.8 : time = 25.34 ms

Ping attempt 2

Reply from 8.8.8.8 : time = 24.87 ms

Ping attempt 3

Reply from 8.8.8.8 : time = 26.12 ms

Ping attempt 4

Reply from 8.8.8.8 : time = 25.56 ms
```

RESULT:

Thus, the Ping program was implemented successfully using Python to check host connectivity and calculate the round-trip time (RTT).